

Experiment no-04

4. Implement a finite-state machine for morphological parsing. In this example, we'll create a simple machine to generate plural forms of English nouns using python.

PROGRAM:

```
def pluralize(noun):
    # Words ending with s, x, z, ch, sh
    if noun.endswith(("s", "x", "z", "ch", "sh")):
        return noun + "es"

    # Words ending with consonant + y
    elif len(noun) > 1 and noun.endswith("y") and noun[-2].lower() not in "aeiou":
        return noun[:-1] + "ies"

    # Default rule
    else:
        return noun + "s"

word = input("Enter a singular noun: ")

print("Plural Form:", pluralize(word))
```

OUTPUT:

```
... Enter a singular noun: book
    Plural Form: books
```